

MANYBERRIES AGCM, AB, Canada

WMO: 710510

Lat: 49.3639N

Lon: 110.6781W

Elev: 3041

StdP: 13.15

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.3	-15.9	-28.7	1.2	-23.1	-20.9	1.9	-15.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.4	89.9	61.2	86.6	60.6	82.7	60.2	64.7	80.6	63.1	79.9	61.6	78.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	85.2	69.1	57.6	79.3	67.8	55.7	73.8	66.2	31.4	80.2	30.1	79.8	29.0	78.6	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-34.0	95.8	6.0	3.0	-38.4	97.9	-41.9	99.7	-45.3	101.3	-49.7	103.5		
(5)			-34.2	67.9	6.0	2.3	-38.6	69.5	-42.1	70.9	-45.5	72.2	-49.8	73.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	15.3	14.4	28.2	40.0	50.5	58.3	65.5	64.4	55.0	40.7	27.8	14.9		
	DBStd	21.37	13.84	15.12	13.67	9.07	7.80	6.60	5.99	6.84	8.84	9.63	12.74	14.00		
	HDD50	5276	1075	997	675	316	91	9	0	1	45	313	667	1087		
	HDD65	9422	1540	1417	1139	750	450	221	67	97	320	753	1116	1552		
	CDD50	1528	0	0	1	17	108	256	480	446	194	25	1	0		
	CDD65	198	0	0	0	0	2	19	82	77	18	0	0	0		
	CDH74	3837	0	0	0	10	123	400	1494	1355	443	12	0	0		
	CDH80	1374	0	0	0	1	16	113	558	526	160	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	15.4	0.8	0.6	0.8	1.4	2.2	3.1	1.5	1.3	1.3	0.9	0.7	0.7		
(15) Precipitation	PrecMax	23.3	1.5	1.5	1.5	2.6	3.7	6.8	5.7	3.5	2.5	2.3	1.7	1.5		
	PrecMin	9.1	0.2	0.1	0.1	0.4	0.4	0.7	0.2	0.3	0.0	0.2	0.2	0.1		
	PrecStd	2.8	0.3	0.4	0.4	0.6	0.9	1.5	1.1	0.8	0.7	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	49.6	63.2	74.1	82.2	90.3	93.6	93.6	90.2	75.5	62.3	48.3		
		MCWB	38.2	41.5	46.9	51.9	57.7	62.3	62.4	61.5	59.4	55.9	47.4	39.9		
	2%	DB	42.4	43.0	57.3	67.9	77.5	82.5	90.1	89.5	85.1	68.0	54.6	42.8		
		MCWB	36.1	36.8	44.0	49.8	56.1	60.6	62.2	60.6	58.7	51.1	43.6	37.1		
	5%	DB	39.2	38.8	51.9	62.4	72.9	78.3	86.3	86.3	78.9	62.5	49.3	38.9		
		MCWB	34.7	34.3	41.4	46.4	53.8	59.0	61.5	60.2	57.0	48.8	40.9	34.5		
	10%	DB	35.4	35.0	46.0	57.4	68.2	74.5	82.5	82.0	73.3	57.6	44.6	34.9		
		MCWB	32.0	32.0	38.4	43.8	52.1	57.4	61.3	59.8	54.8	46.1	37.7	31.5		
	0.4%	WB	39.1	41.9	48.0	53.3	59.2	65.9	68.2	65.4	63.0	56.3	48.6	40.6		
		MCDB	45.8	49.2	61.3	70.3	77.5	78.5	83.3	80.6	80.1	72.7	60.6	46.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.7	37.5	44.9	50.4	57.0	62.9	65.5	63.6	60.3	52.9	44.8	37.0		
		MCDB	41.9	42.2	55.5	67.1	74.7	80.0	81.1	81.1	79.2	64.3	53.3	42.0		
	5%	WB	34.7	34.7	42.0	47.2	55.2	60.3	63.8	62.0	57.8	49.7	41.5	34.6		
		MCDB	39.1	38.4	51.1	60.6	69.3	74.9	79.8	80.3	76.0	60.4	48.5	39.0		
	10%	WB	32.1	32.1	38.7	44.3	53.1	58.4	62.4	60.6	55.6	47.0	38.3	31.5		
		MCDB	35.3	35.2	45.5	55.2	65.1	71.3	78.4	78.2	72.2	57.1	43.7	34.8		
	5% DB	MDBR	20.2	20.7	21.3	24.7	25.5	25.9	29.4	29.1	28.1	25.6	21.6	20.3		
		MCDBR	23.3	24.3	28.8	33.5	34.0	32.0	34.6	35.7	37.2	33.3	28.0	25.0		
		MCWBR	18.2	18.7	17.1	17.6	15.7	14.4	13.4	13.1	15.7	17.9	17.5	19.0		
		MCWBR	22.6	23.4	27.7	32.0	29.5	29.4	29.8	29.9	34.7	30.3	26.3	25.3		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.264	0.289	0.339	0.357	0.350	0.351	0.371	0.327	0.293	0.263	0.249		
		taud	2.421	2.479	2.476	2.338	2.333	2.406	2.418	2.332	2.433	2.517	2.504	2.436		
	Ebn at Noon		257	288	296	288	286	288	286	274	278	272	258	248		
		Edh at Noon	20	23	28	36	38	36	35	36	30	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		409	722	1133	1565	1855	2065	2247	1806	1283	804	457	320		
	RadStd		35	58	83	104	140	121	116	113	103	77	30	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page